

DBMS Lecture 9: Primary Key

Comprehensive Exam Notes for GATE, PSU, University & Technical Interviews

1 1. Introduction & Core Relationship

The concept of **Keys** is a foundational pillar in Database Management Systems (DBMS). It is continuously applied across:

- DBMS Fundamentals & Relational Algebra
- SQL Query Design & Optimization
- Database Design & Normalization (1NF through 5NF/BCNF)
- Entity-Relationship (ER) Model & Relational Mapping
- Competitive Exams (GATE, ISRO, BARC, NET) and Technical Interviews

Key Hierarchy Blueprint

To understand the Primary Key accurately, remember the strict structural reduction hierarchy:

Super Key $\xrightarrow{\text{Remove Redundant Attributes}}$ **Candidate Key** $\xrightarrow{\text{Select Main Identifier}}$ **Primary Key**

2 2. What is a Key?

A **Key** is an attribute or a minimal set of attributes used to uniquely identify a tuple (row) in a relation (table).

Motivation Example

Consider a `Student` relation without a unique identifier:

Name	City	Age
Reddy	Bengaluru	20
Reddy	Bengaluru	20

Using only {Name, City, Age}, it is impossible to determine whether these records represent two distinct students or a duplicate entry. Therefore, a unique key attribute is essential.

3 3. Resolving Ambiguity with Unique Attributes

By introducing a `Roll_No` attribute:

Roll_No	Name	City	Age
1	Reddy	Bengaluru	20
3	Reddy	Bengaluru	20

Now, `Roll_No = 1` unambiguously refers to Student 1, and `Roll_No = 3` refers to Student 2.

4. Why Name Cannot Be a Key

Consider the table:

Roll_No	Name	Age
101	Rahul	20
102	Rahul	21
103	Ravi	20

Multiple real-world entities frequently share identical names. Thus:

- Name → **X Not Guaranteed Unique**
- Roll_No → **✓ Unique**

5. Identification of Candidate Keys

A database table may contain several candidate attributes:

Student(Student_ID, Roll_No, Registration_No, Email, Phone, Aadhaar, PAN)

If the database schema enforces uniqueness and non-nullability for {Roll_No, Registration_No, Email}, then:

Candidate Keys = {{Roll_No}, {Registration_No}, {Email}}

Technical Correction: Real-World Attributes as Keys

Do not automatically assume every real-world unique attribute is a candidate key. For example, two students may share a contact number (e.g., family landline), or phone numbers may be missing. A real-world attribute becomes a candidate key **only if** the relational schema enforces strict uniqueness and required participation constraints.

6. Selection of Primary Key

From the set of available Candidate Keys, the database designer selects **exactly one** to serve as the Primary Key.

Candidate Keys {Roll_No, Registration_No, Email} $\Rightarrow$ Select **Roll_No** $\Rightarrow$ Primary Key = Roll_No

The remaining unselected candidate keys are designated as **Alternate Keys**:

Alternate Keys = {Registration_No, Email}

7. Definition of Primary Key

Formal Definition

A **Primary Key** is a candidate key selected by the database designer to uniquely identify each tuple in a relation.

In relational databases (SQL), the PRIMARY KEY constraint enforces:

PRIMARY KEY = UNIQUE + NOT NULL

8. Key Properties of a Primary Key

1. **Uniqueness Constraint:** No two tuples in a relation can share the same primary key value.
2. **NOT NULL Constraint:** A primary key value cannot be missing, unknown, or undefined (NULL).

Roll_No Value	Status	Reason
101	Valid	Unique integer
102	Valid	Unique integer
101	Invalid	Duplicate value rejected by DBMS
NULL	Invalid	Null value violates Entity Integrity

9. SQL Implementation

Primary keys can be declared during table creation using standard SQL DDL syntax:

```
-- Single Attribute Primary Key
CREATE TABLE Student (
    Roll_No INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Age INT
);
```

10. Why is the Primary Key Essential?

The Primary Key acts as a tuple address for essential relational operations:

- **Retrieval:** SELECT * FROM Student WHERE Roll_No = 101;
- **Modification:** UPDATE Student SET Age = 21 WHERE Roll_No = 101;
- **Deletion:** DELETE FROM Student WHERE Roll_No = 101;

11. Composite Primary Key

Can a Primary Key consist of multiple attributes? **Yes.** When a single attribute is insufficient to guarantee uniqueness, a combination of attributes can form a **Composite Primary Key**.

Example: Enrollment Table

Student_ID	Course_ID	Grade
101	C01	A
101	C02	B
102	C01	A

Here, neither Student_ID nor Course_ID is unique individually. However, the pair (Student_ID, Course_ID) uniquely identifies an enrollment record.

```
-- Composite Primary Key Syntax
CREATE TABLE Enrollment (
    Student_ID INT,
    Course_ID VARCHAR(10),
    Grade CHAR(1),
    PRIMARY KEY (Student_ID, Course_ID)
);
```

```
PRIMARY KEY (Student_ID, Course_ID)
);
```

12 12. Primary Key vs. Candidate Key vs. UNIQUE Constraint

Table 1: Detailed Distinction Matrix

Feature	Primary Key	Candidate Key / UNIQUE
Quantity per Table	Exactly one (1)	Multiple allowed (≥ 0)
NULL Values	Strictly Forbidden (NOT NULL)	Permitted (DBMS dependent; standard SQL allows single or multiple NULLs)
Minimality	Minimal superkey by definition	Minimal superkey by definition
Role	Main identifier for Entity Integrity	Alternative identifier / constraint check
Foreign Key Target	Primary choice for references	Can be referenced by Foreign Key if unique

Golden Statement for Exams

Every Primary Key is a Candidate Key, but every Candidate Key is NOT a Primary Key.

13 13. Primary Key vs. Foreign Key

- **Primary Key (Identity):** Uniquely identifies a record within its own local table.
- **Foreign Key (Relationship):** References a Primary Key (or candidate key) in another table to establish referential integrity.

14 14. Entity Integrity Constraint

Primary Keys directly enforce the **Entity Integrity Rule**:

"No primary key value can be NULL because it is used to identify individual tuples in a relation. A NULL primary key implies an unidentifiable entity, which violates relational model rules."

15 15. Selection Criteria for a Good Primary Key

When choosing among multiple Candidate Keys, select based on:

1. **Uniqueness:** Must strictly guarantee complete tuple uniqueness.
2. **Non-nullability:** Must be available for every possible record.
3. **Stability:** Values should rarely or never change over time.
4. **Simplicity & Efficiency:** Small integer surrogate keys are preferred over long string attributes for indexing speed (B^+ Tree lookups) and join performance.

16 16. Common Pitfalls & Mistakes to Avoid

Exam Caution: Common Misconceptions

- **Mistake 1:** "A table can have two primary keys."
→ **Correction:** A table can have multiple *candidate keys*, but **at most one** PRIMARY KEY constraint.
- **Mistake 2:** "Primary Key must be a single column."
→ **Correction:** It can be a *Composite Primary Key* containing multiple columns.
- **Mistake 3:** "Candidate Keys allow NULL in relational theory."
→ **Correction:** In formal relational theory, candidate keys uniquely identify every tuple and thus do not accept NULL. SQL explicitly mandates NOT NULL for primary keys.

17 17. Target GATE & Interview Q&A

Q1. What is a primary key?

A primary key is the candidate key selected by the database designer to uniquely identify each tuple in a relation.

Q2. How many primary key constraints can exist on a single relation?

At most one primary key constraint per table.

Q3. Can a primary key contain NULL values?

No. Enforced by Entity Integrity and SQL NOT NULL constraints.

Q4. What is a Composite Primary Key?

A primary key that consists of two or more attributes combined to ensure tuple uniqueness.

Q5. What happens to candidate keys that are not selected as the primary key?

They are called **Alternate Keys**.

Q6. Which integrity rule is enforced by the primary key?

Entity Integrity.

Q7. Is every super key a candidate key?

No. A super key may contain redundant attributes. A candidate key is a **minimal** super key.

18 18. One-Minute Revision & Key Formulas

Quick Revision Summary

SUPER KEY $\Rightarrow$ Any set of attributes that uniquely identifies a tuple
CANDIDATE KEY $\Rightarrow$ Minimal Super Key (No redundant attributes)
PRIMARY KEY $\Rightarrow$ Selected Candidate Key (Main identifier)
ALTERNATE KEY $\Rightarrow$ Candidate Keys – Primary Key

Core Formulas to Remember: • PRIMARY KEY = UNIQUE + NOT NULL
• CANDIDATE KEY = MINIMAL + UNIQUE • Primary Key $\subseteq$ Candidate Key $\subseteq$ Super Key

DBMS Challenge Progress: Lecture 9 Complete · Next Topic: Foreign Key & Referential Integrity